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Apparatus for treating substrate and method for aligning dielectric plate using the same

Abstract

Provided is an apparatus for treating a substrate. The apparatus for treating the substrate includes a housing defining a treatment space formed by a combination of an upper housing and a lower housing, a gas supply unit configured to supply gas to the treatment space, a support unit including a chuck configured to support the substrate in the treatment space and an upper electrode provided to surround the check when viewed from a top view, a dielectric plate unit including a dielectric plate arranged to oppose the substrate supported by the support unit in the treatment space, and an upper electrode unit coupled to the dielectric plate unit and including an upper electrode arranged to oppose the lower electrode, in which the upper electrode unit is coupled to the lower housing.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS
(1) This application is a National Stage of International Application No. PCT/KR2021/018052 filed Dec. 2, 2021, claiming priority based on Korean Patent Application No. 10-2021-0115656 filed Aug. 31, 2021, the disclosures of which are incorporated herein by reference in their entireties.
TECHNICAL FIELD
(2) The present disclosure relates to an apparatus for treating a substrate and a method of aligning a dielectric plate, and more particularly, to an apparatus for treating a substrate by using plasma and a method of aligning a dielectric plate provided in the apparatus.
BACKGROUND ART
(3) Plasma is an ionized gas state including ions, radicals, electrons, etc., and is generated in response to very high temperatures, strong electric fields, or radio frequency (RF) electromagnetic fields. A semiconductor device manufacturing process includes an ashing or etching process to remove a film material from a substrate by using plasma. The ashing or etching process is performed by collision or reaction of ions and radical particles contained in plasma with a film material on the substrate. A process of treating a substrate by using plasma may be performed in various manners. A bevel etching apparatus for etching an edge region of the substrate is used to treat the edge region of the substrate by irradiating plasma to the edge region of the substrate.
(4) The bevel etching apparatus performs etching on the substrate by supplying a processing gas excited to a plasma state to the edge of the substrate. To prevent etching of other regions than the edge region of the substrate, a dielectric plate provided to an insulator is located on the substrate. To process the edge region of the substrate, adjustment of a relative position between the dielectric plate and the substrate is crucial.
(5) However, separate inclusion of a structure for moving the dielectric plate causes spatial constraints in forming the interior of an existing bevel etching apparatus and increases the volume of the bevel etching apparatus.
DESCRIPTION OF EMBODIMENTS
Technical Problem
(6) The present disclosure provides an apparatus for treating a substrate and a method of aligning a dielectric plate to efficiently control a relative position between an insulator and the substrate.
(7) The present disclosure also provides an apparatus for treating a substrate and a method of aligning a dielectric plate, by which the efficiency of plasma treatment with respect to an edge region of the substrate may be further improved.
(8) Problems to be solved by the present disclosure are not limited to the above-mentioned problems, and the problems not mentioned may be clearly understood by those of ordinary skill in the art to which the present disclosure belongs from the present specification and the accompanying drawings.

Solution to Problem

(9) The present disclosure provides an apparatus for treating a substrate. The apparatus for treating the substrate includes a housing defining a treatment space formed by a combination of an upper housing and a lower housing, a gas supply unit configured to supply gas to the treatment space, a support unit including a chuck supporting the substrate in the treatment space and an upper electrode provided to surround the check when viewed from a top view, a dielectric plate unit including a dielectric plate arranged to oppose the substrate supported by the support unit in the treatment space, and an upper electrode unit coupled to the dielectric plate unit and including an upper electrode arranged to oppose the lower electrode, in which the upper electrode unit is coupled to the lower housing.

(10) In an embodiment, a step may be provided on an upper portion of the lower housing, and the upper electrode unit may be placed on the step.

(11) In an embodiment, the apparatus may further include a first shape portion formed on a bottom surface of the dielectric plate unit and a second shape portion formed on the support unit to correspond to the first shape portion in a preset position, in which any one of the first shape portion and the second shape portion is provided as a protrusion, and the other is provided as a groove.

(12) In an embodiment, the apparatus may further include a driving member configured to lift and lower the support unit between a lowered position and a lifted position, in which the lowered position is a position where the substrate is carried into the treatment space, and the lifted position is a position where the first shape portion and the second shape portion contact each other.

(13) In an embodiment, the substrate may be carried into the treatment space through a side surface of the housing.

(14) In an embodiment, the substrate may be treated in the treatment space in a treatment position, and the treatment position may be between the lowered position and the lifted position.

(15) In an embodiment, the apparatus may further include a sealing member provided between the upper housing and the lower housing to seal the treatment space.

(16) In an embodiment, the upper electrode unit may further include a second base to which the upper electrode and the dielectric plate unit are coupled.

(17) In an embodiment, the dielectric plate unit may further include a first base arranged between the dielectric plate and the temperature control plate.

(18) In an embodiment, the gas supply unit may include a gas channel formed in a space where the first base and the second base are separated from each other, and a first gas supply unit configured to supply a process gas excited by plasma to the gas channel, and a discharge end of the gas channel may be oriented to an edge region of the substrate supported by the support unit.

(19) In an embodiment, the gas supply unit may include a gas flow path provided in the dielectric plate and a second gas supply unit configured to supply an inert gas to the gas flow path, and the discharge end of the gas flow path may be formed toward a center region of the support supported by the support unit.

(20) The present disclosure also provides a method of aligning a dielectric plate. The method includes aligning the dielectric plate in a preset position with respect to a support unit before or after treating the substrate in a treatment space, in which after the support unit is lifted to a lifted position, the dielectric plate unit may be coupled to a lower housing such that a first shape portion corresponds to a second shape portion.

(21) In an embodiment, when the substrate is carried into the treatment space, the support unit may be placed in a lowered position lower than the lifted position.

(22) In an embodiment, during treatment of the substrate in the treatment space, the support unit may be placed between the lifted position and the lowered position.

Advantageous Effects of Disclosure

(23) According to an embodiment of the present disclosure, a substrate may be efficiently treated.

(24) In addition, according to an embodiment of the present disclosure, plasma treatment may be

uniformly performed on the substrate.
(25) Moreover, according to an embodiment of the present disclosure, the efficiency of plasma treatment on an edge region of the substrate may be further improved.
(26) Effects of the present disclosure are not limited to the above-described effects, and the effects not mentioned may be clearly understood by those of ordinary skill in the art to which the present disclosure belongs from the present specification and the accompanying drawings.

Description

BRIEF DESCRIPTION OF DRAWINGS
(1) FIG. **1** schematically shows substrate treatment equipment according to an embodiment of the present disclosure.
(2) FIG. **2** illustrates an example of an apparatus for treating a substrate provided in a process chamber of FIG. **1**.
(3) FIGS. **3** to **8** each show a state in which a dielectric plate is aligned according to an embodiment of the present disclosure.
MODE OF DISCLOSURE
(4) Hereinafter, examples of the present disclosure will be described in detail with reference to the attached drawings to allow those of ordinary skill in the art to easily carry out the examples of the present disclosure. However, the present disclosure may be implemented in various different forms, and are not limited to the embodiments of the present disclosure described herein. In addition, in describing a preferred embodiment of the present disclosure in detail, when it is determined that a detailed description of a related known function or configuration may unnecessarily obscure the gist of the present disclosure, the detailed description will be omitted. Moreover, the same reference numerals are used for parts having similar functions and actions throughout the drawings.
(5) When a portion is referred to as “includes” a component, the portion may not exclude another component but may further include another component unless stated otherwise. More specifically, it should be understood that the term “include”, “have”, or the like used herein is to indicate the presence of features, numbers, steps, operations, elements, parts, or a combination thereof described in the specifications, and does not preclude the presence or addition of one or more other features, numbers, steps, operations, elements, parts, or a combination thereof.
(6) Singular forms include plural forms unless apparently indicated otherwise contextually. In addition, the shapes and sizes of components in the drawings may be exaggerated for the clearer description.
(7) Hereinafter, an embodiment of the present disclosure will be described in detail with reference to FIGS. **1** and **2**.
(8) FIG. **1** schematically shows substrate treatment equipment according to an embodiment of the present disclosure. Referring to FIG. **1**, substrate treatment equipment **1** may include an equipment front end module (EFEM) **20** and a treatment module **30**. The EFEM **20** and the treatment module **30** may be arranged in one direction.
(9) The EFEM **20** may include a load port **10** and a transfer frame **21**. The load port **10** may be arranged in front of the EFEM **20** in a first direction **11**. The load port **10** may include a plurality of supports **6**. Each support **6** may be arranged in a line in a second direction **12**, and a carrier **4** (e.g., a cassette, front-opening unified pod (FOUP), etc.) having stored therein a substrate W to be provided to a process and a substrate W having been processed may be nested on the support **6**. In the carrier **4**, the substrate W to be provided to the process and the substrate W having been processed are stored. The transfer frame **21** may be between the load port **10** and the treatment module **30**. The transfer frame **21** may include a first transfer robot **25** arranged therein to transfer the substrate W between the load port **10** and the treatment module **30**. The first transfer robot **25**

may transfer the substrate W between the carrier **4** and the treatment module **30** by moving a transfer rail **27** provided in the second direction **12**.
(10) The treatment module **30** may include a load lock chamber **40**, a transfer chamber **50**, and a process chamber **60**. The treatment module **30** may receive the substrate W returned from the EFEM **20** to treat the substrate W.
(11) The load lock chamber **40** may be arranged adjacent to the transfer frame **21**. For example, the load lock chamber **40** may be arranged between the transfer chamber **50** and the EFEM **20**. The load lock chamber **40** may provide a space for allowing the substrate W to be provided to the process to wait before the substrate W is transferred to the process chamber **60**, or allowing the substrate W having been processed to wait before the substrate W is transferred to the EFEM **20**.
(12) The transfer chamber **50** may return the substrate W. The transfer chamber **50** may be arranged adjacent to the load lock chamber **40**. The transfer chamber **50** may have a polygonal body when viewed from a top view. Referring to FIG. **1**, the transfer chamber **50** may have a pentagonal body when viewed from the top view. On the exterior of the body, the load lock chamber **40** and a plurality of process chambers **60** may be arranged along the circumference of the body. A path (not shown) through which the substrate W enters and exits may be formed on each sidewall of the body, and the path may connect the transfer chamber **50** with the load lock chamber **40** or the process chambers **60**. In each path, a door **130** for sealing the interior by opening and closing the path may be provided. In an inner space of the transfer chamber **50**, a second transfer robot **53** may be arranged to transfer the substrate W between the load lock chamber **40** and the process chambers **60**. The second transfer robot **53** may transfer the non-processed substrate W waiting in the load lock chamber **40** to the process chamber **60** or transfer the processed substrate W to the load lock chamber **40**. To sequentially provide the substrate W to the plurality of process chambers **60**, the substrate W may be transferred between the process chambers **60**. When the transfer chamber **50** has a pentagonal body as shown in FIG. **1**, the load lock chamber **40** may be arranged on a sidewall adjacent to the EFEM **20** and the process chambers **60** may be consecutively arranged on the other sidewalls. The transfer chamber **50** may be provided in not only the above-described shape, but also various shapes according to a required processing module.
(13) The process chamber **60** may be arranged adjacent to the transfer chamber **50**. The process chamber **60** may be arranged along the circumference of the transfer chamber **50**. The process chamber **60** may be provided in plural. In each process chamber **60**, processing with respect to the substrate W may be performed. The process chamber **60** may receive the substrate W from the second transfer robot **53** to process the substrate W, and provide the processed substrate W to the second transfer robot **53**. Processing performed in the respective process chambers **60** may be different.
(14) Hereinbelow, an apparatus for treating a substrate to perform plasma processing in the process chamber **60** will be described in detail. Moreover, it will be described below as an example that the apparatus for treating a substrate is configured to perform plasma treatment processing with respect to an edge region of the substrate in the processing chamber **60**. However, without being limited to the example, the apparatus for treating a substrate described below may be equally or similarly applied to various chambers where treatment on the substrate is performed. Moreover, the apparatus for treating a substrate may be equally or similarly applied to various chambers where plasma treatment processing on the substrate is performed.
(15) FIG. **2** illustrates an example of an apparatus for treating a substrate, provided in a process chamber of FIG. **1**. Referring to FIG. **2**, the apparatus for treating a substrate, provided in the process chamber **60**, may perform processing on the substrate W by using plasma. For example, the apparatus for treating a substrate may etch or ash a film material. The film material may include various types of film materials such as a polysilicon film, a silicon oxide film, a silicon nitride film, etc. The film material may also be a natural oxide film or a chemically generated oxide film. In addition, the film material may be a by-product generated during treatment of the substrate W. The

film material may be impurities attached to and/or remaining on the substrate W.

(16) The apparatus for treating a substrate may perform plasma processing on the substrate W. For example, the apparatus for treating a substrate may supply a process gas and generate plasma from the supplied process gas to treat the substrate W. The apparatus for treating a substrate may supply the process gas and generate plasma from the supplied process gas to treat the edge region of the substrate W. Hereinafter, a description will be made using an example where the apparatus for treating a substrate is a bevel etching apparatus that performs etching for the edge region of the substrate W.

(17) The apparatus for treating a substrate may include a housing **100**, a support unit **300**, a dielectric plate unit **500**, an upper electrode unit **600**, a second base **610**, and a gas supply unit **800**.

(18) The housing **100** may include a treatment process **102** therein. An opening **135** may be formed in a surface of the housing **100**. In an example, the opening **135** may be formed in any one of side surfaces of the housing **100**. The substrate W may be carried into or out of the treatment space **102** of the housing **100** through the opening formed in the housing **100**. The opening **135** may be opened and closed by an opening/closing member such as the door **130**.

(19) In an example, the housing **100** may be provided in a form having an open top thereof. The housing **100** may include an upper housing **102** and a lower housing **101**. A treatment space **102** may be formed by a combination of the upper housing **102** and the lower housing **101**. A sealing member **105** may be between the upper housing **102** and the lower housing **101**. The sealing member **105** may seal the treatment space **102**.

(20) When the opening of the housing **100** is opened and closed by the opening/closing member, the treatment space **102** of the housing **100** may be isolated from the outside. In addition, the atmosphere of the treatment space **102** of the housing **100** may be adjusted to low pressure close to vacuum after isolation from the outside. Moreover, the housing **100** may be provided as a material including metal. In addition, the surface of the housing **100** may be coated with an insulating material.

(21) An exhaust hole **104** may be formed in a bottom surface of the housing **100**. Plasma P generated in the treatment space **212** or gases G**1** and G**2** supplied to the treatment space **212** may be exhausted outside through the exhaust hole **104**. Moreover, by-products generated in a process of treating the substrate W by using the plasma P may be exhausted outside through the exhaust hole **104**. The exhaust hole **104** may be connected to an exhaust line (not shown). The exhaust line may be connected to a decompression member providing decompression. The decompression member may provide decompression to the treatment space **102** through the exhaust line.

(22) The support unit **300** may support the substrate W in the treatment space **102**. The support unit **300** may include a chuck **310**, a power source member **320**, an insulating ring **330**, a lower electrode **350**, and a driving member **370**.

(23) The chuck **310** may have a support surface supporting the substrate W. The chuck **310** may have a circular shape when viewed from the top view. The chuck **310** may have a diameter less than the substrate W when viewed from the top view. Thus, the center region of the substrate W supported by the chuck **310** may be nested on the support surface of the chuck **310**, and the edge region of the substrate W may not meet the support surface of the chuck **310**.

(24) A heating means (not shown) may be provided inside the chuck **310**. The heating means (not shown) may heat the chuck **310**. Heating means may be a heater. In addition, a cooling flow path **312** may be formed in the chuck **310**. The cooling flow path **312** may be formed inside the chuck **310**. A cooling fluid supply line **314** and a cooling fluid discharge line **316** may be connected to the cooling flow path **312**. The cooling fluid supply line **314** may be connected to a cooling fluid supply source **318**. The cooling fluid supply source **318** may store cooling fluid and/or the cooling fluid to the cooling fluid supply line **314**. In addition, the cooling fluid supplied to the cooling flow path **312** may be discharged outside through the cooling fluid discharge line **316**. Cooling fluid stored and/or supplied by the cooling fluid supply source **318** may be cooling water or a cooling

gas. The shape of the cooling flow path **312** formed in the chuck **310** may be variously changed without being limited to the shape shown in FIG. **3**. A structure for cooling the chuck **310** may be various structures (e.g., a cooling plate, etc.) capable of cooling the chuck **310** without being limited to a structure for supplying cooling fluid.

(25) The power source member **320** may supply power to the chuck **310**. The power source member **320** may include a power source **322**, a matcher **324**, and a power line **326**. The power source **322** may be a bias power source. The power source **322** may be connected with the chuck **310** through the power source line **326**. The matcher **324** may be provided to the power source line **326** to perform impedance matching.

(26) The insulating ring **330** may be provided to have a ring shape, when viewed from the top view. The insulating ring **330** may be provided surround the chuck **310**, when viewed from the top view. For example, the insulating ring **330** may have a ring shape. The insulating ring **330** may be stepped such that the height of a top surface of an inner region and the height of a top surface of an outer region are different from each other. For example, the insulating ring **330** may be stepped such that the height of the top surface of the inner region is higher than the height of the top surface of the outer region. When the substrate W is nested on the support surface of the chuck **310**, the top surface of the inner region of the insulating ring **330** may contact a bottom surface of the substrate W. When the substrate W is nested on the support surface of the chuck **310**, the top surface of the outer region of the insulating ring **330** may be separated from the bottom surface of the substrate W. The insulating ring **330** may be between the chuck **310** and the lower electrode **350** described below. A bias power source is provided to the chuck **310** such that the insulating ring **330** may be between the chuck **310** and the lower electrode **350** described below. The insulating ring **330** may be provided as an insulating material.

(27) The lower electrode **350** may be arranged under the edge region of the substrate W supported by the chuck **310**. The lower electrode **350** may be provided to have a ring shape, when viewed from the top view. The lower electrode **350** may be provided to surround the insulating ring **330**, when viewed from the top view. The top surface of the lower electrode **350** may be provided at the same height as an outer top surface of the insulating ring **330**. The bottom surface of the lower electrode **350** may be provided at the same height as the bottom surface of the insulating ring **330**. In addition, the top surface of the lower electrode **350** may be provided lower than the top surface of the center portion of the chuck **310**. The lower electrode **350** may be provided spaced apart from the bottom surface of the substrate W supported by the chuck **310**. For example, the lower electrode **350** may be provided spaced apart from the bottom surface of the edge region of the substrate W supported by the chuck **310**.

(28) The lower electrode **350** may be arranged to oppose an upper electrode **620** described later. The lower electrode **350** may be arranged under the upper electrode **620** described later. The lower electrode **350** may be grounded. The lower electrode **350** may increase plasma density by inducing coupling of a bias power source applied to the chuck **310**. Thus, the efficiency of treatment on the edge region of the substrate W may be improved.

(29) The driving member **370** may lift the chuck **310**. The driving member **370** may include a driver **372** and an axis **374**. The axis **374** may be coupled to the chuck **310**. The axis **374** may be connected with the driver **372**. The driver **372** may lift the chuck **310** in an up-down direction through the axis **374**. In an example, the driving member **370** may lift and lower the chuck **310** between a lowered position, a treatment position, and a lifted position. In an example, the lowered position may be a position of the chuck **310** when the substrate W is carried into the treatment space **102**, the treatment position may be a position of the chuck **310** when the substrate W is treated in the treatment space **102**, and the lifted position may be a position of the chuck **310** when the dielectric plate **520** is aligned in the horizontal direction. In an example, the treatment position may be higher than the lowered position, and the lifted position may be higher than the treatment position.

(30) The dielectric plate unit **500** may include a dielectric plate **520** and a first base **510**. In addition, the dielectric plate unit **500** may be coupled to the second base **610** described later.
(31) The dielectric plate **520** may have a circular shape when viewed from the top view. In an example, the bottom surface of the dielectric plate **520** may be provided in a flat shape. The dielectric plate **520** may be arranged to oppose the substrate W supported by the support unit **300** in the treatment space **102**. For example, the dielectric plate **520** may be arranged on the support unit **300**. The dielectric plate **520** may be provided as a material including ceramic. In the dielectric plate **520**, a gas flow path connected to a first gas supply unit **810** of the gas supply unit **800** described below may be formed. In addition, a discharge end of the gas flow path may be structured such that the first gas G**1** supplied by the first gas supply unit **810** may be supplied to the center region of the substrate W supported by the support unit **300**. The discharge end of the gas flow path may also be structured such that the first gas G**1** is supplied to the top surface of the center region of the substrate W supported by the support unit **300**.
(32) The first base **510** may be arranged between the dielectric plate **520** and the second base **610** described later. The first base **510** may be coupled to the second base **610** described below, and the dielectric plate **520** may be coupled to the first base **510**. Thus, the dielectric plate **520** may be coupled to the second base **610** through the first base **510**.
(33) In an example, the top surface of the first base **510** may have a less diameter than the bottom surface of the dielectric plate **520**. The top surface of the first base **510** may have a flat shape. The bottom surface of the first base **510** may have a stepped shape. For example, the bottom surface of the edge region of the first base **510** may be stepped such that the height of the bottom surface of the edge region is less than the height of the bottom surface of the center region of the first base **510**. The bottom surface of the first base **510** and the top surface of the dielectric plate **520** may have shapes combinable with each other. In an example, the first base **510** may be provided as a material including metal. For example, the first base **510** may be provided as a material including aluminum.
(34) The upper electrode unit **600** may include a second base **610** and an upper electrode **620**. In addition, the upper electrode unit **600** may be coupled to the second base **610** described later. Thus, the dielectric plate unit **500** and the upper electrode unit **600** may be coupled to each other through the second base **610**.
(35) In an example, the upper electrode unit **600** may be coupled to the lower housing **101**. For example, an upper portion of the lower housing **101** may be stepped, and the upper electrode unit **600** may be placed on a step of the lower housing **101**. In an example, the upper electrode **620** may be placed on the step of the lower housing **101**. Selectively, the second base **610** may be placed on the step of the lower housing **101**. An outer diameter of the upper electrode **620** placed on the step of the lower housing **101** may be less than a diameter of the step of the lower housing **101**. Thus, in a method of aligning the dielectric plate **520** described later, the dielectric plate **520** may be moved to a position where it is horizontally aligned.
(36) The upper electrode **620** may oppose the above-described lower electrode **350**. The upper electrode **620** may be arranged on the lower electrode **350**. The upper electrode **620** may be arranged on the edge region of the substrate W supported by the chuck **310**. The upper electrode **620** may be grounded.
(37) The upper electrode **620** may have a shape surrounding the dielectric plate **520**, when viewed from the top view. In an example, the upper electrode **620** may be provided in the shape of a ring. The upper electrode **620** may be separated from the dielectric plate **520** to form a separating space. The separating space may form a part of a gas channel in which the second gas G**2** supplied by the second gas supply unit **830** described below flows. The discharge end of the gas channel may be structured such that the second gas G**2** may be supplied to the edge region of the substrate W supported by the support unit **300**. The discharge end of the gas channel may also be structured such that the second gas G**2** may be supplied to the top surface of the edge region of the substrate

W supported by the support unit **300**.

(38) The second base **610** may have a ring shape, when viewed from the top view. The top surface and the bottom surface of the second base **610** may have a flat shape. The second base **610** may be provided to be separated from the first base **510**. The second base **610** may be separated from the first base **510** to form a separating space. The separating space may form a part of a gas channel in which the second gas G**2** supplied by the second gas supply unit **830** described below flows. In an example, the second base **610** may be provided as a material including metal. For example, the second base **610** may be provided as a material including aluminum.

(39) In an example, the second base **610** may be provided as a temperature control plate capable of generating heat. For example, the second base **610** may generate hot or cold heat. The second base **610** may generate hot heat or cold heat to control the temperatures of the dielectric plate unit **500** and the upper electrode unit **600** to be maintained relatively constant. For example, the second base **610** may generate cold heat to suppress the temperatures of the dielectric plate unit **500** and the upper electrode unit **600** to the maximum so as not to be excessively high in a process of treating the substrate W.

(40) The gas supply unit **800** may supply gas to the treatment space **102**. The gas supply unit **800** may supply the first gas G**1** and the second gas G**2** to the treatment space **102**. The gas supply unit **800** may include a first gas supply unit **810** and a second gas supply unit **830**.

(41) The first gas supply unit **810** may supply the first gas G**1** to the treatment space **102**. The first gas G**1** may be an inert gas such as nitrogen, etc. The first gas supply unit **810** may supply the first gas G**1** to the center region of the substrate W supported by the chuck **310**. The first gas supply unit **810** may include a first gas supply source **812**, a first gas supply line **814**, and a first valve **816**. The first gas supply source **812** may store the first gas G**1** and/or supply the first gas G**1** to the first gas supply line **814**. The first gas supply line **814** may be connected with a flow path formed on the dielectric plate **520**. The first valve **816** may be installed in the first gas supply line **814**. The first valve **816** may be an on/off valve or a flow control valve. The first gas G**1** supplied by the first gas supply source **812** may be supplied to the center region of the top surface of the substrate W through the flow path formed on the dielectric plate **520**.

(42) The second gas supply unit **830** may supply the second gas G**2** to the treatment space **102**. The second gas G**2** may be a process gas that is excited to a plasma state. The second gas supply unit **830** may supply the second gas G**2** to the edge region of the substrate W through a gas channel formed by separation between the dielectric plate **520**, the first base **510**, the upper electrode **620**, and the second base **610**, provided on the edge region of the substrate W supported by the chuck **310**. The second gas supply unit **830** may include a second gas supply source **832**, a second gas supply line **834**, and a second valve **836**. The second gas supply source **832** may store the second gas G**2** and/or supply the second gas G**2** to the second gas supply line **834**. The second gas supply line **814** may supply the second gas G**2** to a separating space functioning as a gas channel. The second valve **836** may be installed in the second gas supply line **834**. The second valve **836** may be an on/off valve or a flow control valve. The second gas G**2** supplied by the second gas supply source **832** may be supplied to the edge region of the top surface of the substrate W through the second flow path **602**.

(43) In an embodiment, the apparatus for treating a substrate may be provided with a first shape portion **530** and a second shape portion **315**. The first shape portion **530** and the second shape portion **315** may be provided as corresponding shapes in a preset position. In an example, the first shape portion **530** may be formed on a bottom surface of the dielectric plate unit **500**, and the second shape portion **315** may be provided on the support unit **300**. Any one of the first shape portion **530** and the second shape portion **315** may be provided as a protrusion, and the other may be provided as a groove. For example, the protrusion **530** may be formed on the bottom surface of the dielectric plate unit **500**, and the groove **315** may be provided on the top surface of the support unit **300**. In an example, the preset position may be a position where the support unit **300** is placed

in the lifted position. When the support unit **300** is placed in the lifted position, the first shape portion **530** and the second shape portion **315** may be provided to contact each other. For example, when the support unit **300** is placed in the lifted position, the protrusion **530** may be inserted into the groove **315**.

(44) Hereinbelow, a method of aligning the dielectric plate **520** according to the present disclosure will be described with reference to FIGS. **3** to **8**.

(45) In an example, the dielectric plate **520** may be aligned in a preset position with respect to the support unit **300** before or after the substrate W is treated in the treatment space **102**. Before or after the substrate W is processed in the treatment space **102**, the dielectric plate **520** may be aligned in a state where the substrate W is not provided in the treatment space **102**. In an example, treatment of the substrate W may be bevel etching that treats the edge region of the substrate W.

(46) First, as shown in FIG. **3**, before or after the substrate W is treated in the treatment space **102**, the upper housing **102** may be opened and the dielectric plate unit **500** or the upper electrode unit **600** may be removed from the housing **100**, for maintenance of the dielectric plate unit **500** or the upper electrode unit **600**. The bottom surface of the chuck **310** may be in a position spaced apart from the bottom of the lower housing **101** by a first distance D.sub.1.

(47) Before the housing **100** is sealed, the chuck **310** may be moved to the lifted position as shown in FIG. **4**. In the lifted position, the bottom surface of the chuck **310** may be in a position spaced apart from the bottom of the lower housing **101** by a second distance D.sub.2. In an example, the second distance D.sub.2 may be provided as a distance farther than the first distance D.sub.1.

(48) The operator may couple the dielectric plate unit **500** and the upper electrode unit **600** with the lower housing **101** such that the protrusion **530** provided in the dielectric plate **520** and the groove **315** provided in the support unit **300** correspond to each other, in the lifted position as shown in FIG. **5**. For example, the operator may place the second base **610** to which the dielectric plate unit **500** and the upper electrode unit **600** are coupled on the lower housing **101**. At this time, the upper electrode **600** may be stopped by the step of the lower housing **101**. In addition, the dielectric plate unit **500** and the upper electrode unit **600** may be coupled to the lower housing **101** such that the protrusion **530** may be inserted into the groove **315**. The position of the dielectric plate unit **500** may serve as an important factor for etching the edge region of the substrate W. In particular, when the dielectric plate **520** is not at the right position, the edge region of the substrate W may be unbalancedly etched with respect to the center of the substrate W. Thus, it is necessary to align the dielectric plate **520** at the right position with respect to the center of the substrate W. However, the substrate W is not in the treatment space **102** during non-treatment of the substrate W, such that the horizontal position of the dielectric plate **520** may be aligned with respect to the support unit **300** on which the substrate W is placed.

(49) Thereafter, as shown in FIG. **6**, the support unit **300** may be lowered to the lowered position. In an example, in the lowered position, the bottom surface of the chuck **310** may be in a position spaced apart from the bottom of the lower housing **101** by a third distance D.sub.3. In an example, the third distance D.sub.3 may be provided as a distance closer than the first distance D.sub.1. Thereafter, as shown in FIG. **7**, the substrate W may be carried into the treatment space **102** through the opening. When the substrate W is placed on the support unit **300**, the substrate W may be bevel-etched as shown in FIG. **8**. During treatment of the substrate W, the support unit **300** may be placed in the treatment position. In an example, in the treatment position, the bottom surface of the chuck **310** may be in a position spaced apart from the bottom of the lower housing **101** by the first distance D.sub.1.

(50) According to the present disclosure, when the dielectric plate is positioned on the substrate to treat the edge region of the substrate, the horizontal position of the dielectric plate **520** may be aligned.

(51) According to the present disclosure, the dielectric plate **520** may be coupled to the lower housing **101**, such that the position of the dielectric plate **520** may not be changed even when the

upper housing **102** is opened after the horizontal position of the dielectric plate **520** is aligned.
(52) While an example is described where the apparatus for treating the substrate performs etching with respect to the edge region of the substrate W, the present disclosure is not limited thereto. The foregoing embodiments may be equally or similarly applied to various facilities and processing requiring treatment of the edge region of the substrate W.
(53) A method of generating the plasma P by the apparatus for treating the substrate described in the foregoing example may be an inductive coupled plasma (ICP) scheme. The above-described method of generating the plasma P by the apparatus for treating the substrate may be a capacitor couple plasma (CCP) scheme. The apparatus for treating the substrate may generate the plasma P by using both the ICP scheme and the CCP scheme or a scheme selected from the ICP scheme and the CCP scheme. The apparatus for treating the substrate may treat the edge region of the substrate W through a known method of generating the plasma P.
(54) The above detailed description exemplifies the present disclosure. In addition, the foregoing description is provided by showing a preferred embodiment of the present disclosure, and the present disclosure may be used in various combinations, changes and environments. That is, it is possible to change or modify within the scope of the concept of the present disclosure disclosed herein and the equivalent range to the foregoing disclosure and/or the range of the technique or knowledge of the art. The above-described embodiment describes the best state for implementing the technical idea of the present disclosure, and may also make various changes required in the specific application field and use of the present disclosure. Therefore, the detailed description of the present disclosure is not intended to limit the present disclosure to the disclosed embodiment. In addition, the accompanying claims should be interpreted as including other embodiments.

Claims

1. An apparatus for treating a substrate, the apparatus comprising: a housing defining a treatment space formed by a combination of an upper housing and a lower housing; a gas supply unit configured to supply a gas to the treatment space; a support unit comprising a chuck supporting the substrate in the treatment space and a lower electrode provided to surround the chuck when viewed from a top view; a dielectric plate unit comprising a dielectric plate arranged to oppose the substrate supported by the support unit in the treatment space; and an upper electrode unit coupled to the dielectric plate unit and comprising an upper electrode arranged to oppose the lower electrode; a first shape portion formed on a bottom surface of the dielectric plate unit; and a second shape portion formed on the support unit to correspond to the first shape portion in a preset position, wherein any one of the first shape portion and the second shape portion is provided as a protrusion, and the other is provided as a groove, and wherein the upper electrode unit is coupled to the lower housing.
2. The apparatus of claim 1, wherein a step is provided on an upper portion of the lower housing, and the upper electrode unit is placed on the step.
3. The apparatus of claim 1, further comprising a driving member configured to lift and lower the support unit between a lowered position and a lifted position, wherein the lowered position is a position where the substrate is carried into the treatment space, and the lifted position is a position where the first shape portion and the second shape portion contact each other.
4. The apparatus of claim 3, wherein the substrate is carried into the treatment space through a side surface of the housing.
5. The apparatus of claim 3, wherein the substrate is treated in the treatment space in a treatment position, and the treatment position is between the lowered position and the lifted position.
6. The apparatus of claim 1, further comprising a sealing member provided between the upper housing and the lower housing to seal the treatment space.
7. The apparatus of claim 1, wherein the upper electrode unit further comprises a second base to

which the upper electrode and the dielectric plate unit are coupled.
8. The apparatus of claim 7, wherein the dielectric plate unit further comprises a first base arranged between the dielectric plate and a temperature control plate.
9. The apparatus of claim 8, wherein the gas supply unit comprises: a gas channel formed in a space where the first base and the second base are separated from each other; and a first gas supply unit configured to supply a process gas excited by plasma to the gas channel, and a discharge end of the gas channel is oriented toward an edge region of the substrate supported by the support unit.
10. The apparatus of claim 8, wherein the gas supply unit comprises: a gas flow path provided in the dielectric plate; and a second gas supply unit configured to supply an inert gas to the gas flow path, and the discharge end of the gas flow path is formed toward a center region of the support supported by the support unit.
11. A method of aligning the dielectric plate using the apparatus of treating the substrate of claim 1, the method comprising aligning the dielectric plate in a preset position with respect to the support unit before or after treating the substrate in the treatment space, wherein after the support unit is lifted to a lifted position, the dielectric plate unit is coupled to the lower housing such that the first shape portion corresponds to the second shape portion.
12. The method of claim 11, wherein when the substrate is carried into the treatment space, the support unit is placed in a lowered position lower than the lifted position.
13. The method of claim 12, wherein during treatment of the substrate in the treatment space, the support unit is placed between the lifted position and the lowered position.